GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Digital Circuits

Paper Code: EC • Section 5: Digital Circuits (8-10 Marks)

Subject Code: EC-DIG • **Target Exam:** GATE ECE • **Weightage:** 8-10 Marks

1. Combinational Circuits & Multiplexer Implementations

- **Universal MUX:** A 2^n:1 multiplexer implements any n-variable function directly, or (n+1)-variable function using basic inverters.
- **Full Adder:** Sum = A oplus B oplus C_in; Carry = AB + C_in(A oplus B). Requires 2 Half Adders and 1 OR gate.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Full Adder Logic	**Sum = A oplus B oplus C_in | C_out = AB + C_in(A oplus B)**	A, B, C_in = Addend Bits	Standard building block of ripple and lookahead adders
Max Clock Frequency	**f_{max} = 1 / (t_{cq} + t_{comb} + t_{setup})**	t_cq, t_comb, t_setup = Delays	Guarantees zero setup-time violation in synchronous register paths

GATE PYQ (2 Marks) - Maximum Clock Frequency of Synchronous Counter
Question: Flip-flops have t_cq = 12 ns, t_setup = 5 ns. Combinational feedback has delay 18 ns. Find max clock frequency.

Step 1: T_clk >= t_cq + t_comb + t_setup = 12 + 18 + 5 = 35 ns.
Step 2: f_max = 1 / (35 * 10^-9) = **28.57 MHz**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** Watch out for race-around conditions in level-triggered JK flip-flops when J=K=1 and pulse width > t_cq.
- **Important:** In K-maps, group sizes must always be powers of 2.